

Emma Butler

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Neuroscience MSc graduate with a highly analytical, detail-oriented mindset and direct experience in large-scale data processing and analysis, custom Python scripting, and cross-functional workflow creation in healthcare and research. Skilled in numerous wet lab techniques, histology, microscopy methods, and multidisciplinary group collaboration. Seeking a research role in a pharmaceutical or biotech environment. Willing to relocate at my own expense.

Education

University of St Andrews – MSc by Research, Neuroscience <i>St Andrews, Scotland</i>	9/2023 – 12/2025
Washington & Lee University – BS, Neuroscience <i>Lexington, VA</i>	9/2019 – 5/2023
Virginia Western Community College – AS, Health Science <i>Roanoke, VA</i>	8/2015 – 6/2019

Technical Skills

- **Academic Research** – lab animal research, scientific collaboration, wet lab experience, human postmortem and iPSC tissue handling, conducting *in vivo* electrophysiology experiments (BIOL 360 Experimental Neurophysiology - Washington & Lee), genotype testing, IF antibody staining, microplate assays, confocal and fluorescence microscopy, Zeiss Airyscan
- **Data Analysis & Programming** – large data processing, data workflow automation, software integration, Python (Pandas, NumPy, SciPy), FIJI/ImageJ, image binarization and quantification, data cleaning and wrangling, exploratory data analysis, reproducible data pipelines for biological datasets, statistical analysis (SPSS, Excel), data-driven decision making, SQL (basics)
- **Visualization & Reporting** – communication, collaboration, presentations and public speaking, detailed research documentation, Python visualizations (Matplotlib, Seaborn), 3D modelling (FIJI, Imaris for Neuroscientists), publication graphics

Academic Experience

University of St Andrews – Postgraduate Research Intern | Neuroscience (ALS/MNDs) 9/2023 – 7/2025

- Conducted ALS research using a novel TDP43ΔNLS mouse model under the supervision of Prof Gareth Miles and Dr Matthew Broadhead in the Neural Control of Movement Lab
- European Journal of Neuroscience: “Hallmarks of Synaptopathy in a Mouse Model of Sporadic Amyotrophic Lateral Sclerosis” by Ani Avayzian-Hancock, Emma Butler, Claire Meehan, Gareth Miles, Matthew Broadhead
- Presented my research findings to the School of Psychology & Neuroscience, participated in the school’s annual postgraduate poster session, and invited to attend the 2024 Scottish Neuroscience Group seminar at the University of Aberdeen
- Shadowed colleagues during *ex vivo* murine experiments and live electrophysiological recordings
- Skilled in fluorescent antibody staining procedures, multiplexing methodology development, human and murine postmortem tissue handling as well as human iPSCs, qPCR and blotting procedures, numerous microscopy techniques, and image analysis
- Designed and implemented custom FIJI macros and Python pipelines to process and analyze 1000+ super-resolution synaptic imaging datasets, reducing manual analysis time by ~85%
- Produced data visualizations and insights to drive research decisions and support scientific communication

Washington and Lee University – Undergraduate Research Intern | Biochemistry 9/2021 – 5/2023

- Mammalian cell culture basic training
- Developed effective *in vitro* methodology for measuring ribosome processivity in mRNA samples treated with TORIN1
- Quantified protein translation at incremental time points using luciferase assays and the Infinite M1000 Pro Tecan Instrument
- Ran Kolmogorov-Smirnov and ANOVA tests in Excel for processivity differences in treated samples

Professional Experience

DataAnnotation – Contract AI Trainer | Remote, USA 5/2026 - present

- Training AI models for real-world applications

Nashville Urology PC – Clinical Data Specialist & Lead Medical Scribe | Nashville, TN 4/2025 – 4/2026

- Ensured accurate and HIPAA-compliant EMR documentation in real time and conducted structured clinical data entry to support billing and clinical reporting processes based on regulatory requirements
- Implemented procedural and documentation improvements for providers by utilizing diagnostic/treatment pathway expertise and physician collaboration, leading to noticeable enhancements in reporting accuracy, operational efficiency, and billing workflows
- Assisted with sample testing, clinical data analysis, billing audits, maintaining CPT/ICD-10 coding standards, and training new hires

Charleston Gastroenterology Associates – Patient Care Coordinator | Charleston, WV 05/2022 – 09/2022

- Coordinated scheduling and referrals, triaged patients, collected vitals, updated EMR data, tested lab samples